Course Handout (Mobile Application Development)
(Batch: 2024-2028) - (Even Sem - 2026)

Course Code: CSE3709	**Course Name:** Mobile Application Development	
Credits: 2 (1 L – 0 D – 2 P)	**Contact hours per week:** One hour theory and two hours lab	
Batch: 2024-2028, 4th Sem **Academic Year:** 2025-26	**Semester Duration:** 19th Jan 2026 to 8th May 2026	
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Aim of the course:

This course introduces students to programming technologies, design and development related to mobile applications. Topics include accessing device capabilities, industry standards, operating systems and programming for mobile applications using an OS Software Development Kit (SDK). Upon completion, students should be able to create basic applications for mobile devices.

Course Outcomes (CO):

At the end of the course the students should be able to do the following:

	Course Outcome Statement	Mapped Component	Associated Activities / Assessment
CO1	Understand the Android OS and its underlying architecture, and Android Studio.	Employability (Understanding)	→ Install Android Studio and configure SDK tools on your system. → Open Android Studio, show project structure (Manifest, Java/Kotlin files, XML layouts). → Demonstrate the creation of a basic "Hello World" app in Android Studio and explain the project structure.
CO2	Apply different views and layout resources for a given real-life scenario.	Skill Development (Apply)	→ Hands-on practical sessions in labs/lectures. → Instructor provides incomplete XML layout; students complete it using proper views (TextView, ImageView, Button, etc.). → Students sketch the layout on paper first, then implement using different layout managers. → Share a broken layout file; students fix alignment and constraints.
CO3	Create and develop Android applications and run them over real devices and Android virtual devices (AVDs).	Entrepreneurship, Skill Development (Creating & Evaluating)	→ Students build a simple apps in the class based on the topics covered. → Pair students; one runs the app on an emulator, the other on a physical device. → Instructor introduces common errors (e.g., missing permissions); students troubleshoot.

Prerequisites, if any:

1. Java Programming Language
2. Computer Fundamentals

Topics of Course:

Topics	Number of Sessions (32)	Course Outcomes
Introduction to Android & Development Environment: Android Foundation – Evolution of Mobile Platforms, Android Ecosystem, Android Versions, Android Architecture, DVM & ART; Development Environment - Java and Android Studio IDE Installation, Android project structure (Gradle, manifests, resources), SDK Tools and AVD Manager, Different Android Studio Tools, Emulator vs real-device testing.	2 (1 Th + 1 Lab)	CO1
Activities, Activity Lifecycle: Introduction to Activity - Creating an Activity, Activity Lifecycle; Running the Application in the AVD, stopping a Running Application, Reviewing the Resource Files and other project files.	2 (1 Th + 1 Lab)	CO1
Intents: Intent Overview, Implicit Intents (Activity Navigation), Explicit Intents (Sharing, Calling, opening URLs), Intents with Activities, Intent Filters, Passing Data Between Activities.	3 (2 Th + 1 Lab)	CO2, CO3
Android UI Design (XML + Java) and Event Handling: View & ViewGroup hierarchy, Layouts: ConstraintLayout (primary), LinearLayout, RelativeLayout (basic understanding), WebView Layout; Material Design principles; Views: TextView, EditText, Buttons, ImageView, ProgressBar, Switch, ToggleButton, CheckBox, RadioGroup; RecyclerView (detailed): ViewHolder pattern, Adapters, Click listeners; Cards & Material Components, Toast, Snackbar, Dialogs, AlertDialog, Styles & Themes; Menus: Options menu, Context menu, Popup menu; Dialogs: AlertDialog, Custom dialogs.	8 (4 Th + 4 Lab)	CO2, CO3
Working with Device Hardware: Camera integration, Camera Intent, Capturing images & video; Sensors: Accelerometer, proximity, light, SensorManager, SensorEventListener; Audio & video playback: MediaPlayer, ExoPlayer (basic introduction).	4 (2 Th + 2 Lab)	CO2, CO3
Networking & APIs: Internet permissions & security, REST API concepts, JSON parsing, Making API calls using, HttpURLConnection (basic), Retrofit; Authentication using Google Firebase.	4 (2 Th + 2 Lab)	CO2, CO3
Data Storage: Directory Structure and Saving Data in Files, SQLite database: Tables, CRUD operations.	4 (2 Th + 2 Lab)	CO2, CO3
Mobile Communication Services: Sending SMS, sending email, sending notifications, Dialling a number	2 (1 Th + 1 Lab)	CO2, CO3
Location-Based Services: Introduction to location-based service, configuring the Android Emulator for Location Based Services, Geocoding and Map-Based Activities.	3 (1 Th + 2 Lab)	CO2, CO3
Total Contact Sessions:	**32 Sessions (16 Th + 16 Lab)** *(Total Contact Hours: 16+2x16 = 48 Hrs)*	

CO-PO Mapping:

Course (Level) and Course Outcome Mapping to Program Outcomes *[Three Levels of Mapping: 1: means weak, 2: means medium and 3: means strong]*

Course (Level) Mapping PO1		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
Mapping of Course Outcomes	CO1	2	2														
	CO2	3	3	1	1	1						2	1		1		
	CO3	3	3	1	1	1					2	2	1		1		

Learning Resources:

Textbooks:

1. John Horton, Android Programming for Beginners, Third Edition, Packt Publishing, 2021
2. Neil Smyth, "Android Studio 4.1 Development Essentials - Java Edition", First Edition, Payload Media Inc., 2020

Reference Links:

1. https://developer.android.com/courses

Experiential Learning: 80%

Assessment Pattern: The final grade will be based on the marks/ grades obtained in the Project evaluations along with the quiz and assignment. Relative grading method defined in the academic regulations of the university will be followed to grade the students. Student must secure minimum 40% of marks after completing all the assessments in the following table to become eligible for grading.

Component	Weightage	Evaluation Schedule/Timeline	Rubrics/Remark	CO Mapping
Quiz	30% (1 x 30 = 30 marks)	Before Mid-Sem Exam	There will be one quiz in Offline mode. The quiz will have 30 multiple choice questions.	CO1, CO2
Project Evaluation Phase 1	30% (1 x 30 = 30 marks)	In the week of Mid-Sem Exam	It will have two parts: - Assignment (15 marks): It would have few very small Android applications to implement. - In phase-1 (15 Marks): The students will bring the Figma designs of the app, and present the concept of the Android app.	CO1, CO2, CO3
End Term (Project Evaluation Phase 2)	40% (1 x 40 = 40 marks)	Last week of April	The students will show the complete running project (app), presentation, and viva-voce.	CO1, CO2, CO3

Student Responsibilities:

- Attend lectures and do labs as per instructions.
- Prepare for discussions on analysis part on lab work during defined class every week.

Attendance Policy: Students are expected to attend the classes regularly. Failure to attend the classes regularly and adhere to the expected attendance percentage.
Lab: Lab work will have an analysis part and will be discussed in the defined class in every week. Based on the performance on analysis and discussion part, the marks will be given. Late submission in assignment is not allowed, and any late submission will be awarded “0” marks in that assignment.
Recourse examination policy: As per University policy.
Make-up policy: No make-up work will be given for unexcused absences. The faculty needs to be informed in advance in case the student is not going to be able to submit an assignment or take any evolution component, and it is at the discretion of the faculty to sanction make up for an evaluation component.
Academic dishonesty/cheating/plagiarism: Plagiarism and dishonesty in any form in any evaluation component will lead to appropriate disciplinary action.